MATH 3603–Homework 3

Due: 29-March-2026

1. Let X and Y be independent geometric random variables with parameters α and β. Compute the probability mass function of $X + Y$.

2. Find the probability density function of $X + Y$ when X and Y have joint probability density function

$$f(x,y) = \begin{cases} \frac{1}{2}(x+y)e^{-(x+y)}, & x, y \geq 0 \\ \\ 0, & \text{otherwise} \end{cases}$$

3. Let X and Y have joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{x}, & 0 \leq y \leq x \leq 1 \\ \\ 0, & \text{otherwise} \end{cases}$$

Compute (i). $f_X(x)$; (ii). $f_{Y|X}(y|x)$; (iii). $\mathbb{P}(X^2 + Y^2 \leq 1 | X = x)$; (iv). $\mathbb{P}(X^2 + Y^2 \leq 1)$.

4. Let N be Poisson distributed with parameter λ. Show that, for any function g such that the expectations exist,

$$\mathbb{E}(Ng(N)) = \lambda \mathbb{E}g(N+1).$$

More generally, if

$$S = \sum_{r=1}^{N} X_r,$$

where $\{X_r : r \geq 0\}$ are independent identically distributed non-negative integer-valued random variables and they are also independent of N, show that

$$\mathbb{E}(Sg(S)) = \lambda \mathbb{E}(g(S + X_0)X_0).$$

Note that in the above equation X_0 is independent of S.

5. Show that the conditional expectation $\psi(X) = \mathbb{E}(Y|X)$ satisfies $\mathbb{E}(\psi(X)g(X)) = \mathbb{E}(Yg(X))$ for any function g for which both expectations exist.